QUANTUMFLOOR

Computational Research Report

Job ID: c024a245

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'label'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	100
Unique Rows:	100
Duplicate Rows:	0
Total Columns:	5
Missing Values:	0.00%
Leakage Risk Detected:	0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	5
Quantum Embedding Dimension:	2
Qubit Budget:	2
Classical Training Size:	80
Quantum Subset Size:	80

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.3158	0.0000	0.2632	
logreg	Classical	0.1053	0.0000	0.1053	
svm_rbf	Classical	0.1579	0.0000	0.1579	
random_forest	Classical	0.3947	0.0000	0.3158	
qsvm	Quantum	0.5263	0.0000	0.5263	
pqk	Quantum	0.3684	0.0000	0.3684	
vqc	Quantum	0.1579	0.0000	0.0000	

7. Comparison & Explainability

Best Classical AUROC: 0.3947

Best Quantum AUROC: 0.5263

Delta: 0.1316

Explainability Status: available

PC2: 0.0468

PC1: 0.0427

8. Final Verdict

Best Overall Model: qsvm

Quantum Advantage: Yes